New Document

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§1 lecture_8

§1.1 Split exact sequence

Recall that we can construct a new group G if H, K are normal subgroups. The construction is simply given by cartesian product.

This gives us a trivial exact sequence:

$$1 \to K \xrightarrow{i} K \times H \xrightarrow{\pi} H \to 1$$

Where i is given by the usual inclusion map and π by projection. However, what if H isn't normal?

Definition 1.1 (split exact sequence)**.** A exact sequence $1 \to K \xrightarrow{i} G \xrightarrow{\pi} H \to 1$ is called split when there exist homomorphism $\varphi : H \to G$ such that $\pi \circ \varphi = \mathrm{id}_H$

We will show that we can form "semi-direct product" for H and K if and only we can make the above exact sequence split.

§1.2 Group extension; semi-direct product

Usually, we will have K to be a kernel of homomorphism and $H = G/K$ to be the associated quotient group. However, most of the time $G \neq G/K \times K$, this happens already when $D_6 \neq D_6/C_3 \times C_3 = C_2 \times C_3$.

Definition 1.2 (group extension)**.** If (Q) and (N) are two groups, then (G) is an extension of (Q) by (N) if there is a short exact sequence

$$1 \to N \xrightarrow{\iota} G \xrightarrow{\pi} Q \to 1.$$

A way that it is useful is if N is the maximal normal subgroup of G then the quotient group Q must be simple.

Definition 1.3 (semi-direct product)**.** Given a map $\varphi : H \to \mathrm{Aut}(N)$, we can construct the group semi-direct product: $G = N \times_\varphi H$ or $N \rtimes H$ by taking the set of G as $N \times H$ and the operation be $(n_1, h_1)(n_2, h_2) = (n_1(h_1 \cdot n_2), h_1 h_2)$ where the left action on N is given by the automorphism map.

Example 1.4 (semi-direct product)**.**

- We have $D_{2n} = C_n \rtimes C_2$. It is obvious that $C_2 \to \mathrm{Aut}(C_n)$ has $s \mapsto (r \mapsto r^{-1})$.
- The trivial automorphism gives the direct product
- $S_n = A_n \rtimes C_2$ The intuition is an automorphism of A_n is by changing index of the permutation. Thus we define the automorphism by conjugation.
  I believe everything work fine if we have $\varphi((a(12)b(12), e)) = \varphi((a, (12)))\varphi(b, (12)) = a(12)b(12)$

in S_n

Definition 1.5 (split exact sequence)**.** It is obvious that the above example all gives a split exact sequence. So here we will have example of non-split sequence.

- $1 \to C_2 \to C_4 \to C_2 \to 1$
- $1 \to C_2 \to \mathrm{SU}(2) \to \mathrm{SO}(3) \to 1$

§1.3 Classification result

We first classify groups of order pq where $p < q$. There are two cases, p even or p odd.

§1.4 $|G| = 2q$

We consider a 2-sylow H_2 and a q-sylow subgroup H_q of G. Note that $n_q = 1$ since n_q divides 2. Thus H_q is normal. However, $n_2 = 1 \bmod 2 \Rightarrow 1 \vee q$. If $n_2 = 1$ then it is trival.

So suppose $n_2 = q$. Note that we have the exact sequence:

$$1 \to C_q \xrightarrow{i} G \xrightarrow{\pi} C_2 \to 1$$

Consider $C_2 \mapsto \mathrm{Aut}(C_q) = C_{q-1}$. The automorphism mapped must be +- . So the group is D_{2q} by Definition 1.3

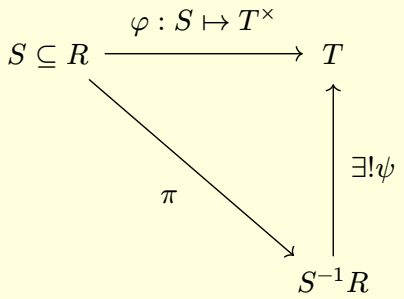
§2 Lecture 10

§2.1 Localization

We try to generalize field of fraction to non-integral domain

- equivalence relation $(r_1, s_1) \sim (r_2, s_2)$

Definition 2.1 (Universal Property)**.** Let S be a multiplicative closed set, then consider the homomorphism $R \xrightarrow{\varphi} T$, it maps S to the set of units in T.



How do we construct such $S^{-1}R$? For example, the zero divisors in $\mathbb{Z}/6\mathbb{Z}$ are $\{2,3,4\}$, if we want $2,4$ to be invertible, then we actually need 3 to be 0 otherwise the units we want will be zero divisors.

The equivalence relation is defined by $(r_1, s_1) \sim (r_2, s_2)$ if there exists s' such that

$$s'(r_1 s_2 - r_2 s_1) = 0$$

Note the $\ker \pi = S^0$, so $S^{-}1R = \frac{R}{S^0}$

Example 2.2 (Localization)**.**

- $S = \{1,2,4\}, R = \mathbb{Z}/6\mathbb{Z}$, then $S^{-1}R = (\mathbb{Z}/6\mathbb{Z})/(3\mathbb{Z}) = \mathbb{Z}/3\mathbb{Z}$

In ring $\mathbb{Z}$,

- $S = \mathbb{Z} - \{0\}$, then the localization is $\mathbb{Q}$
- $S = \{1, 6, 6^2, ...\}$, then $S^{-}1R = \mathbb{Z}\left[\frac{1}{6}\right]$, (we can check everything is invertible)
- $S = \mathbb{Z} - (p)$??? $\text{Spec}\mathbb{Z}_{(p)} = \{0, (2), (3), ..., (p)\}$
- $k[x], (x-a)$ is a maximal ideal, localize with $k[x] - (x-a) = \left\{\frac{f(x)}{g(x)} \mid (x-a) \nmid g(x)\right\}$

§2.2 Localization of modules

This is just changing the set of scalars. We can define $S^{-1}M = \left\{\frac{x}{s} : x \in M, s \in S\right\}$. So basically the set of scalers is changed to $S^{-1}R$

$$S^{-1}M \cong S^{-1}R \otimes_R M$$

§2.3 Diagram chasing

- five lemma
- snake lemma

???